VEHICLE INSPECTION REPORT

AI-Powered Body + Engine Sound Analysis · June 06, 2026 at 09:00

Physical Defect Detection (7-class YOLO + Severity Model) + Engine Knock Audio Classification

VIN / Registration	JTHBK1GG7R1234567	**Mileage**	45000
Make / Model	LEXUS LEXUS-SUV-TEST	**Year**	2024

AI Vehicle Health Assessment

48/100

Vehicle Health Score

Risk Level	**High**	**Overall Status**	**FAIL**

Key Risk Factors:

- Safety-critical components affected: Light — Lamp Crack, Glass Crack
- High-confidence damage on: Light — Lamp Crack, Fender — Scratch, Scratch
- Minor cosmetic damage on: Door — Scratch, Part Crack

AI Inspection Summary

The inspection detected: Light — Lamp Crack, Door — Scratch, Fender — Scratch, Scratch, Part Crack, Glass Crack. Safety-critical damage identified on Light — Lamp Crack, Glass Crack requires immediate attention.

Defect Analysis — Dual Model Results

Light — Lamp Crack ■ Safety-Critical (Severe, 80.2% confidence)
Damaged lights reduce road legality — replace before driving at night.

Door — Scratch (Moderate, 65.1% confidence)
Inspect door seals and hinges — repair to prevent water ingress.

Fender — Scratch (Severe, 88.7% confidence)
Exposed metal will oxidise — treat and repair at nearest opportunity.

Door — Scratch (Minor, 48.7% confidence)
Inspect door seals and hinges — repair to prevent water ingress.

Scratch (Moderate, 76.8% confidence)
Scratches expose bare metal — treat with touch-up paint to prevent rust.

Part Crack (Moderate, 74.8% confidence)
Consult a qualified technician for assessment and repair.

Part Crack (Moderate, 69.8% confidence)
Consult a qualified technician for assessment and repair.

Glass Crack ■ Safety-Critical (Moderate, 60.1% confidence)
Glass cracks compromise structural integrity and visibility — replace immediately.

AI Recommendations

1. Damaged lights reduce road legality — replace before driving at night.
2. Inspect door seals and hinges — repair to prevent water ingress.
3. Exposed metal will oxidise — treat and repair at nearest opportunity.
4. Scratches expose bare metal — treat with touch-up paint to prevent rust.
5. Cracked component detected — assess structural impact and repair.

Standard 7-Part Damage Checklist

Component	Defect Type / Severity	Confidence	Status
Bonnet	No Damage Detected	—	—
Bumper	No Damage Detected	—	—
Dickey	No Damage Detected	—	—
Door	Door — Scratch	65.1%	Moderate
Fender	Fender — Scratch	88.7%	Severe
Light	Light — Lamp Crack	80.2%	Severe ■
Windshield	No Damage Detected	—	—
Total Unique Parts/Types	**6**	**Overall Status:**	**FAIL**

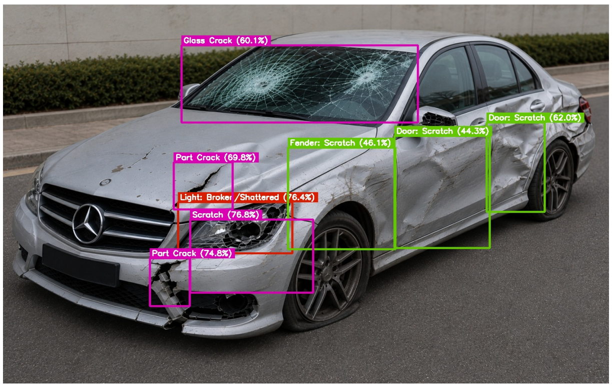
Additional Damage Detections (Severity Model)

Damage Type	Confidence	Severity Tier
Scratch	76.8%	Moderate
Part Crack	74.8%	Moderate
Part Crack	69.8%	Moderate

Glass Crack	60.1%	Moderate

Annotated Images — Dual Model Detections

Blue/green boxes = part detections; Purple boxes = severity-only detections



This report was generated using the unified MEHRA inspection pipeline: physical defect detection with a 7-class part model and severity model, engine sound knock classification, and Groq LLaMA 3.3 for report analysis. Results should be verified by a qualified technician.